

COMPUTER STUDIES CLASS VIII

Time Allowed: 2 Hours | Chapters Covered: 1 to 5 (Balanced) | Maximum Marks: 80

General Instructions:

- Answers to this paper must be written on the answer script provided separately.
- You will not be allowed to write during the first 15 minutes. This time is to be spent in reading the question paper.
- Attempt all questions from Section A (40 Marks).
- Attempt all five questions from Section B (40 Marks).
- The intended marks for questions or parts of questions are given in brackets [].

SECTION A (40 MARKS)

Question 1

[10 × 1 = 10]

Choose the most appropriate option for each of the following questions:

- (a) Which function of an Operating System ensures that multiple running programs receive adequate CPU time without freezing the system?
(i) File Management (ii) Processor Management (iii) Device Driver Setup (iv) Power Management
- (b) Which Operating System interface requires users to type text-based commands to execute tasks?
(i) GUI (Graphical User Interface) (ii) CUI (Character User Interface)
(iii) Touchscreen Interface (iv) Icon Interface
- (c) Which symbol must precede any formula or function in MS Excel?
(i) # (ii) = (iii) @ (iv) &
- (d) In MS Excel, what does the reference \$C\$4 represent?
(i) Relative Reference (ii) Absolute Reference (iii) Mixed Reference (iv) Circular Reference
- (e) Which chart is circular and divided into slices to represent proportional data of a whole?
(i) Bar Chart (ii) Pie Chart (iii) Line Chart (iv) Radar Chart
- (f) Which element in a chart identifies what each color, pattern, or symbol represents?
(i) Legend (ii) Gridline (iii) Plot Area (iv) Category Axis
- (g) Which flowchart shape represents an Input or Output operation?
(i) Rectangle (ii) Parallelogram (iii) Oval (iv) Rhombus
- (h) A step-by-step procedure written in simple human language to solve a specific problem is called:
(i) Flowchart (ii) Algorithm (iii) Program Code (iv) Compiler
- (i) The mechanism of binding data variables and methods together into a single unit (like a medicine capsule) is known as:
(i) Polymorphism (ii) Encapsulation (iii) Inheritance (iv) Abstraction
- (j) Which Java operator calculates and returns the remainder after integer division?
(i) / (Slash) (ii) % (Modulus) (iii) * (Asterisk) (iv) == (Comparison)

Question 2

- (a) State any two reasons why an Operating System is required to act as an intermediary between user software and hardware. [2]
- (b) Identify whether each of the following cell addresses is Relative, Absolute, or Mixed:
(i) \$D5 (ii) B8 (iii) \$K\$12 [3]
- (c) Name the four fundamental pillars of Object-Oriented Programming (OOP). [2]

- (d) Distinguish between the Chart Area and the Plot Area in a spreadsheet chart. [3]

Question 3

- (a) Evaluate the result of each of the following Java expressions: [2]
(i) $18 - 4 \times 3 + 6$ (ii) $29 \% 5$
- (b) What does the error ##### mean when seen in an Excel worksheet cell? Mention how a user can fix it. [2]
- (c) Name the appropriate flowchart symbol used for: [2]
(i) Starting or Stopping an algorithm (ii) Checking whether a condition is True or False
- (d) Differentiate between an Object and a Class in Java with a simple real-world analogy. [3]
- (e) Identify the most appropriate Java primitive data type to store: [3]
(i) The roll number of a student (e.g., 24) (ii) The average marks of a student (e.g., 87.5)

SECTION B (40 MARKS)

Answer all five questions. Each question carries 8 Marks.

Question 4

- (a) Explain any two major functions of an Operating System from the following: [3]
(i) Memory Management (ii) File Management (iii) Device Management
- (b) Differentiate between a GUI (Graphical User Interface) and a CUI (Character User Interface) based on: [3]
(i) User friendliness for beginners (ii) System resource (RAM/CPU) consumption
- (c) Give one practical real-life application where a Real-Time Operating System (RTOS) is required instead of a standard desktop OS. Give one reason why. [2]

Question 5

- (a) Observe the sample worksheet given below and answer the following questions: [3]

	A	B	C	D
1	Student Name	Computer	Science	Total Marks
2	Aarav	88	82	
3	Meera	95	90	
4	Karan	72	68	

Write the Excel formula to:

- (i) Calculate the Total Marks of Aarav in cell D2.
(ii) Find the highest score obtained in Computer marks in cell B5.
(iii) Calculate the average score in Science in cell C5.
- (b) Differentiate between a Formula and a Function in MS Excel with one example of each. [3]
- (c) Briefly explain what causes the following errors in an Excel sheet: [2]
(i) #DIV/0! (ii) #NAME?

Question 6

- (a) What is a chart in MS Excel? State two advantages of visualizing numeric data through charts rather than tables. [3]

- (b) Suggest the most suitable chart type for each of the following scenarios: [3]
(i) Showing hourly temperature variations in your city over a 24-hour period.
(ii) Representing the percentage contribution of different budget heads to an annual total.
(iii) Comparing the quarterly sales figures of four different regional branches.
- (c) Differentiate between the Category Axis (X-axis) and the Value Axis (Y-axis) of a column chart. [2]

Question 7

- (a) Draw neat and labelled diagrams for each of the following flowchart symbols: [3]
(i) Terminal Symbol (ii) Processing Box (iii) Decision Box
- (b) Write a clear, step-by-step algorithm to input two numbers from the user, find the larger of the two, and display the result. [3]
- (c) What is the purpose of a connector in a flowchart? When is it used? [2]

Question 8

- (a) Define and explain the following two pillars of Object-Oriented Programming with a simple daily-life example for each: [3]
(i) Data Abstraction (ii) Inheritance
- (b) Differentiate between: [3]
(i) Encapsulation and Data Abstraction
(ii) Unary Operator and Binary Operator
- (c) Write the Java expressions corresponding to the following mathematical statements: [2]
(i) $d = b^2 - 4ac$
(ii) $\text{Area} = \text{length} \times \text{breadth}$

— End of Question Paper —